Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

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TE - 1	Batch: T11	Sem: 5	Roll No:12	Date:12/9/2026

Experiment 07

Aim: To implement monitoring using Prometheus / Grafana / Nagios

LO Mapping: LO5

LO5: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

THEORY:

1. Continuous Monitoring in IT Infrastructure

Continuous monitoring is a systematic operational process designed to observe, measure, detect, report, and respond to performance degradation, application failures, and security threats across an organization's IT infrastructure. Once an application or machine learning pipeline is deployed into production, continuous monitoring ensures operational availability, resource stability, and proactive threat response.

2. Key Monitoring Tools

- Nagios: An open-source monitoring and alerting system designed to track applications, services, operating systems, and network protocols. It enables administrators to identify and resolve infrastructure issues before they disrupt business processes.
- Prometheus: An open-source metrics-based monitoring and alerting toolkit built primarily for multi-dimensional time-series data collection using a pull model over HTTP.
- Grafana: An open-source visualization and analytics software that integrates with database and monitoring engines (like Prometheus or Nagios) to display metrics via customizable dashboards.

3. Remote Monitoring & Protocol Terms

- NRPE (Nagios Remote Plugin Executor): A Nagios daemon that allows a central Nagios server to remotely execute plugins (like disk usage, CPU load, or memory checks) on remote Linux/Unix hosts.
- NSClient++: A Windows-native monitoring agent and service that acts as the Windows equivalent of NRPE. It runs in the background on Windows hosts, listens for requests **from**

the Nagios server over the NRPE protocol, executes local system checks, and returns the results to Nagios.

- PING Check: An ICMP (Internet Control Message Protocol) status check used by Nagios to measure network round-trip time and packet loss, verifying whether a target host machine is online and reachable.

4. Containerized Deployment Model

- WSL2 (Windows Subsystem for Linux 2): A lightweight utility in Windows that runs an actual Linux kernel inside a managed virtual machine, enabling Linux-native software to run on Windows without performance overhead.
- Docker Desktop: A platform that uses WSL2 to run containerized Linux applications (such as Nagios Core) natively on a Windows host system.

Implementation:

Step 1: Pull and Run Nagios Container
Executed in Administrator Command Prompt:

docker run --name nagios4 -d -p 8080:80 jasonrivers/nagios:latest

Step 2: Configure Web Authentication Password
Created the login credentials for user nagiosadmin:

docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin

(Entered password: 123…)

Step 3: Download and Install NSClient++ Agent
Downloaded the NSClient++ MSI installer using TLS 1.2 protocol in PowerShell and executed silent installation:

powershell -Command "[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12; Invoke-WebRequest -Uri 'https://github.com/mickem/nscp/releases/download/0.5.2.35/NSCP-0.5.2.35-x64.msi' -OutFile 'NSCP.msi'"

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powershell -Command "Start-Process msiexec.exe -ArgumentList '/i NSCP.msi ADDLOCAL=ALL MODULES=CheckExternalScripts,NRPEServer /qn' -Wait"

Step 4: Configure NSClient++ (nsclient.ini) & Restart Service
Edited C:\Program Files\NSClient++\nsclient.ini to set allowed hosts:

Ini, TOML

allowed hosts = 127.0.0.1

Restarted the NSClient++ service:

powershell -Command "Restart-Service nscp"

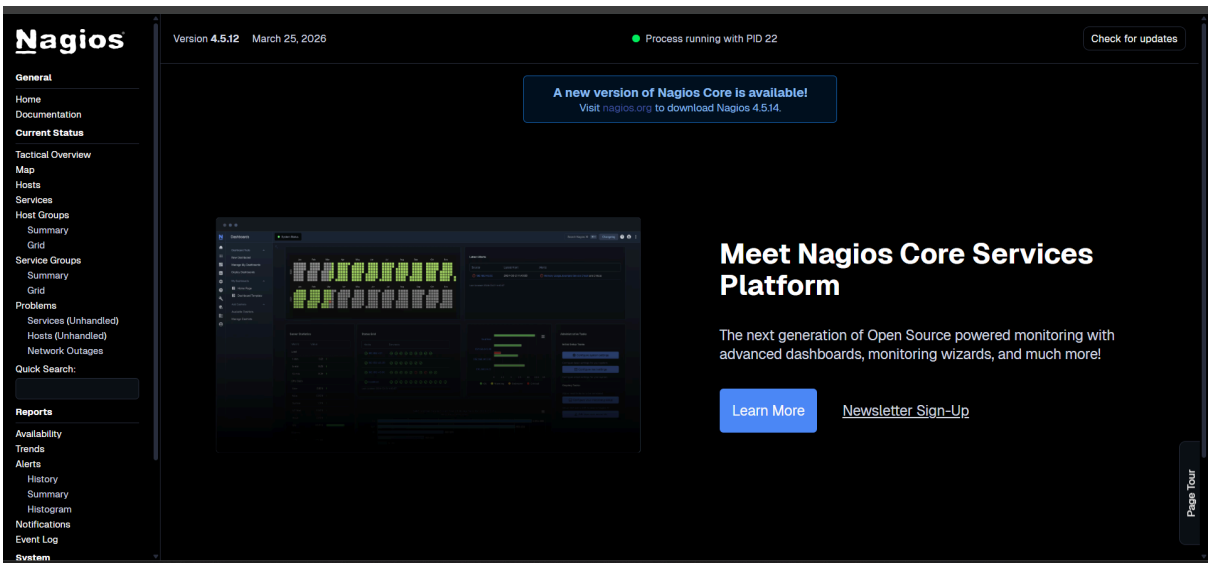
Step 5: Verify Host in Nagios Dashboard

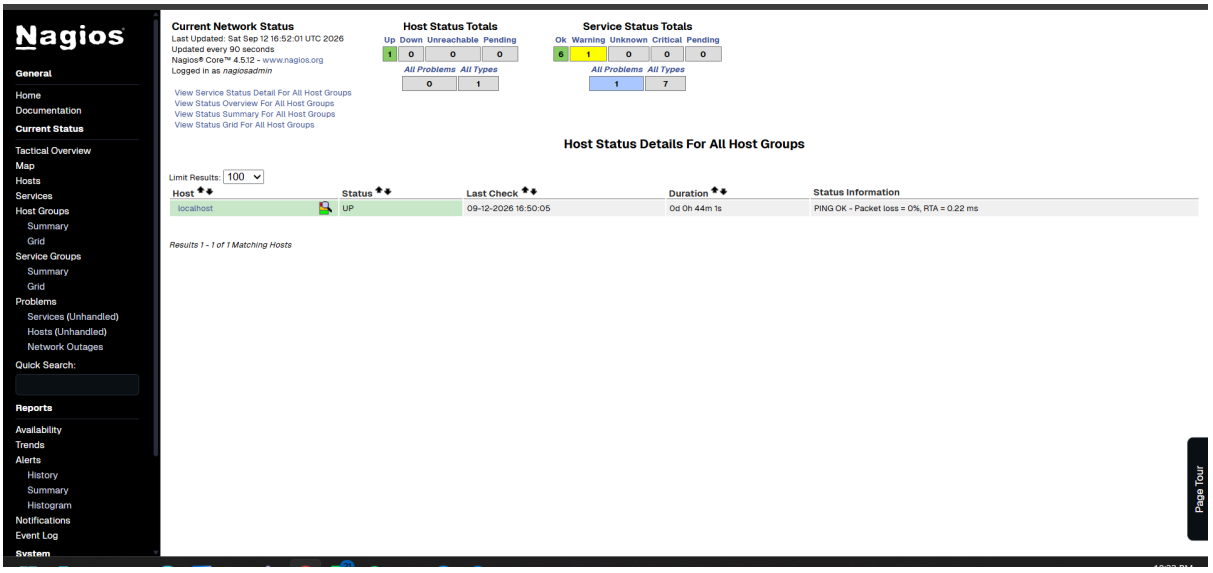
1. Opened browser at http://localhost:8080/nagios/.
2. Logged in using username nagiosadmin and password 12….
3. Navigated to Current Status rightarrow Hosts and Services.

Output:

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Conclusion:

In this experiment, continuous infrastructure monitoring was successfully implemented using Nagios Core and NSClient++. A Linux-native Nagios Core instance was deployed on Windows using Docker Desktop backed by WSL2. The NSClient++ agent was configured on the Windows host to enable remote metric execution via the NRPE protocol. Active host monitoring and connectivity checks (PING) were configured and verified through the Nagios Web Interface dashboard